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In[1]:= (* =====*)
(*Verification of low Mach number expansion for u and varphi up to 8th order*)
(*All equations should yield {0,0} or 0.)(*If all pass,
prints "True" and total computation time.*/)
(* =====*)
(*-----Define base quantities-----*)
(*-----0th order-----*)
u0 = { (3 Tf) / (4 (3 μ + α)) (2 μ + α - α x2^2), 0};

(*-----2nd order-----*)
φ2 = - ( (3 Tf (-2 + x1) α μ) / (2 γ (α + 3 μ))) ;

u2 = { (9 Tf^2 α (Tf x2^4 α ((-5 + x2^2) α - 10 μ) + 40 (-2 + x1) μ (α - x2^2 α + 2 μ) (α + 3 μ))) /
(320 γ (α + 3 μ)^3), 0};

(*-----4th order-----*)
φ4 =
- ( (9 Tf^2 α μ (3 Tf x1 (α + 2 μ)^2 + 4 (α + 3 μ) (α (-5 + x2^2 + (-2 + x1)^2 γ) μ + (-1 + x2^2)
α ν - 2 μ (μ + ν)))) / (32 γ^2 (α + 3 μ)^3) );

u4 = {1 / (17920 γ^2 (α + 3 μ)^5) × 9 Tf^3 α
(12 Tf x1 α^2 γ μ (α + 3 μ) ((6 + 7 x2^4 (-5 + x2^2)) α + 14 (1 - 5 x2^4) μ) + 84 (-2 + x1) α μ
(α + 3 μ) (Tf x2^4 α ((-5 + x2^2) α - 10 μ) + 40 (-2 + x1) μ (α - x2^2 α + 2 μ) (α + 3 μ)) -
1 / 20 x2^4 α (3360 Tf α γ ((-5 + x2^2) α - 10 μ) μ (α + 3 μ) +
Tf^2 (α^3 (-90 (-35 + γ) + 210 x2^2 (-2 + γ) - 45 x2^4 (3 + 2 γ) + 2 x2^6 (7 + 5 γ)) -
30 α^2 (7 (-90 + γ) - 28 x2^2 (-2 + γ) + x2^4 (9 + 6 γ)) μ + 840 α
(45 + x2^2 (-2 + γ)) μ^2 + 25200 μ^3) - 5600 α μ (α + 3 μ)^2 ((8 + γ) μ + 6 ν)) +
420 μ (α - x2^2 α + 2 μ) (α + 3 μ) (3 Tf x1 (α + 2 μ)^2 + 4 (α + 3 μ)
(α (-5 + x2^2 + (-2 + x1)^2 γ) μ + (-1 + x2^2) α ν - 2 μ (μ + ν)))) ,
- ( (27 Tf^4 x2 α^3 μ ((6 - 7 x2^4 + x2^6) α - 14 (-1 + x2^4) μ)) / (4480 γ (α + 3 μ)^4) ) };

(*-----6th order-----*)
φ6 = -1 / (17920 γ^3 (α + 3 μ)^5) × 9 Tf^3 α μ
(9 Tf^2 x1 (α + 2 μ) (3 α^3 (35 + 2 γ) + 14 α^2 (45 + γ) μ + 1260 α μ^2 + 840 μ^3) +
1120 α μ (α + 3 μ)^2 (α (-2 - 6 x2^2 + 3 x1 (-5 + x2^2) + 36 γ + x1 (-9 + (-6 + x1) x1) γ +
6 (-2 + x1) x2^2 γ + 2 (-2 + x1)^3 γ^2) μ - 6 (-2 - 4 γ + x1 (5 + 3 γ)) μ^2 +
3 α (2 - 3 x1 - 2 x2^2 + x1 x2^2 + 2 (-2 + x1) (-1 + x2^2) γ) ν -
6 (-2 + 3 x1 + 2 (-2 + x1) γ) μ ν) + 84 Tf (α + 3 μ)
(-120 μ^3 (μ + ν) + 60 α μ^2 ((-3 + x2^2 + x1 (-8 + 3 x1) γ) μ + (-3 + x2^2) ν) +
10 α^2 μ ((-9 + 6 x2^2 + x2^4 - (6 (8 - 3 x1) x1 + x2^4) γ) μ + (-9 + 6 x2^2 + x2^4) ν) +
α^3 ((-15 + 15 x2^2 + x2^4 (-5 + x2^2) (-1 + γ) + 15 x1 (-8 + 3 x1) γ) μ -
(15 - 15 x2^2 - 5 x2^4 + x2^6) ν))) ;

u6 = {1 / (11480268800 γ^3 (α + 3 μ)^7) × 27 Tf^4 α
(3843840 Tf x1^2 α^3 (2 + 2 γ + γ^2) μ^2 (α + 3 μ)^2 ((6 - 35 x2^4 + 7 x2^6) α +
14 (1 - 5 x2^4) μ) + 3363360 α μ (α + 3 μ) (Tf x2^4 α ((-5 + x2^2) α - 10 μ) +
40 (-2 + x1) μ (α - x2^2 α + 2 μ) (α + 3 μ)) (3 Tf x1 (α + 2 μ)^2 +
4 (α + 3 μ) (α (-5 + x2^2 + (-2 + x1)^2 γ) μ + (-1 + x2^2) α ν - 2 μ (μ + ν))) -
1456 x1 α^2 μ (α + 3 μ) (10560 Tf (-1 + x2^2) α (2 + γ (2 + γ)) μ
(α + 3 μ) ((-6 - 6 x2^2 + x2^4) α - 14 (1 + x2^2) μ) -

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$$\begin{aligned}
& \text{Tf}^2 (\alpha^3 (12545 + 4 \gamma (4765 + 16 \gamma) + 55 x2^8 (17 + 4 \gamma (5 + \gamma)) + 198 \\
& \quad x2^4 (-66 + \gamma (-103 + 2 \gamma)) - 330 x2^6 (1 + \gamma (-1 + 2 \gamma)) - 2 x2^{10} \\
& \quad (41 + 2 \gamma (24 + 5 \gamma))) + 22 \alpha^2 (3671 + 5468 \gamma + 58 \gamma^2 + 42 x2^4 (-88 + \\
& \quad (-134 + \gamma) \gamma) + 5 x2^8 (17 + 4 \gamma (5 + \gamma)) - 60 x2^6 (1 + \gamma (-1 + 2 \gamma))) \mu - \\
& \quad 1320 \alpha (-127 - 2 \gamma (94 + \gamma) + 63 x2^4 (2 + 3 \gamma) + x2^6 (1 + \gamma (-1 + 2 \gamma))) \mu^2 - \\
& \quad 55440 (-1 + x2^4) (2 + 3 \gamma) \mu^3) - \\
& \quad 24640 (-1 + x2^4) \alpha (1 + \gamma) \mu (\alpha + 3 \mu)^2 ((8 + \gamma) \mu + 6 \nu) + \\
& \quad x2 (42240 \text{Tf} x2 (-1 + x2^2) \alpha (2 + 2 \gamma + \gamma^2) ((-3 + x2^2) \alpha - 7 \mu) \mu (\alpha + 3 \mu) + \\
& \quad 21120 \text{Tf} x2 \alpha (2 + \gamma (2 + \gamma)) \mu (\alpha + 3 \mu) \\
& \quad ((-6 - 6 x2^2 + x2^4) \alpha - 14 (1 + x2^2) \mu) - 4 \text{Tf}^2 x2^3 \\
& \quad (-5 x2^6 \alpha^3 (41 + 48 \gamma + 10 \gamma^2) + 110 x2^4 \alpha^2 (17 + 20 \gamma + 4 \gamma^2) (\alpha + 2 \mu) - \\
& \quad 495 x2^2 \alpha (1 - \gamma + 2 \gamma^2) (\alpha + 2 \mu)^2 + 66 (3 \alpha^3 (-66 - 103 \gamma + 2 \gamma^2) + \\
& \quad 14 \alpha^2 (-88 - 134 \gamma + \gamma^2) \mu - 1260 \alpha (2 + 3 \gamma) \mu^2 - 840 (2 + 3 \gamma) \mu^3)) - \\
& \quad 98560 x2^3 \alpha (1 + \gamma) \mu (\alpha + 3 \mu)^2 ((8 + \gamma) \mu + 6 \nu)) + \\
& \quad 160160 \mu (\alpha - x2^2 \alpha + 2 \mu) (\alpha + 3 \mu) (9 \text{Tf}^2 x1 (\alpha + 2 \mu) \\
& \quad (3 \alpha^3 (35 + 2 \gamma) + 14 \alpha^2 (45 + \gamma) \mu + 1260 \alpha \mu^2 + 840 \mu^3) + \\
& \quad 1120 \alpha \mu (\alpha + 3 \mu)^2 (\alpha (-2 - 6 x2^2 + 3 x1 (-5 + x2^2) + 36 \gamma + x1 (-9 + (-6 + x1) x1) \\
& \quad \gamma + 6 (-2 + x1) x2^2 \gamma + 2 (-2 + x1)^3 \gamma^2) \mu - 6 (-2 - 4 \gamma + x1 (5 + 3 \gamma)) \mu^2 + \\
& \quad 3 \alpha (2 - 3 x1 - 2 x2^2 + x1 x2^2 + 2 (-2 + x1) (-1 + x2^2) \gamma) \nu - \\
& \quad 6 (-2 + 3 x1 + 2 (-2 + x1) \gamma) \mu \nu) + 84 \text{Tf} (\alpha + 3 \mu) (-120 \mu^3 (\mu + \nu) + \\
& \quad 60 \alpha \mu^2 ((-3 + x2^2 + x1 (-8 + 3 x1) \gamma) \mu + (-3 + x2^2) \nu) + 10 \alpha^2 \mu \\
& \quad ((-9 + 6 x2^2 + x2^4 - (6 (8 - 3 x1) x1 + x2^4) \gamma) \mu + (-9 + 6 x2^2 + x2^4) \nu) + \\
& \quad \alpha^3 ((-15 + 15 x2^2 + x2^4 (-5 + x2^2) (-1 + \gamma) + 15 x1 (-8 + 3 x1) \gamma) \\
& \quad \mu - (15 - 15 x2^2 - 5 x2^4 + x2^6) \nu))) + \\
& \quad x2^4 \alpha (32032 \text{Tf}^2 \alpha \mu (\alpha + 3 \mu) (\alpha^3 (-90 (-31 + 2 (-33 + \gamma) \gamma) + 420 x2^2 \\
& \quad (2 + \gamma + \gamma^2) - 45 x2^4 (17 + 4 \gamma (5 + \gamma)) + x2^6 (82 + 4 \gamma (24 + 5 \gamma))) - \\
& \quad 30 \alpha^2 (14 (-43 + (-88 + \gamma) \gamma) - 56 x2^2 (2 + \gamma + \gamma^2) + 3 x2^4 (17 + 4 \gamma (5 + \gamma))) \\
& \quad \mu + 840 \alpha (45 + 90 \gamma + 2 x2^2 (2 + \gamma + \gamma^2)) \mu^2 + 25200 (1 + 2 \gamma) \mu^3) + \\
& \quad \text{Tf}^3 (\alpha^5 (-91 x2^8 (15 + 8 \gamma) (93 + 20 \gamma) + 91 (-195355 + 7180 \gamma + 64 \gamma^2) - \\
& \quad 1287 x2^4 (-3009 - 832 \gamma + 88 \gamma^2) + 12 x2^{10} (754 + 575 \gamma + 90 \gamma^2) + \\
& \quad 36036 x2^2 (-101 + \gamma (-89 + 2 \gamma)) + 2002 x2^6 (9 + 8 \gamma (13 + 4 \gamma))) - \\
& \quad 182 \alpha^4 (999119 - 22528 \gamma - 638 \gamma^2 + x2^8 (15 + 8 \gamma) (93 + 20 \gamma) + \\
& \quad 132 x2^2 (1234 + (1156 - 13 \gamma) \gamma) - 44 x2^6 \\
& \quad (9 + 8 \gamma (13 + 4 \gamma)) + 99 x2^4 (-1299 + \gamma (-377 + 33 \gamma))) \mu + \\
& \quad 4004 \alpha^3 (84 x2^2 (-268 + (-262 + \gamma) \gamma) - 135 x2^4 (-87 + 2 (-13 + \gamma) \gamma) + \\
& \quad 30 (-6173 + 2 \gamma (34 + \gamma)) + 2 x2^6 (9 + 8 \gamma (13 + 4 \gamma))) \mu^2 - \\
& \quad 360360 \alpha^2 (-14 (-298 + \gamma) + 336 x2^2 (1 + \gamma) + x2^4 (-87 + 2 (-13 + \gamma) \gamma)) \mu^3 - \\
& \quad 60540480 \alpha (25 + x2^2 (1 + \gamma)) \mu^4 - 605404800 \mu^5) - \\
& \quad 358758400 \alpha^2 (1 + \gamma) \mu^2 (\alpha + 3 \mu)^3 ((8 + \gamma) \mu + 6 \nu) - \\
& \quad 160160 \text{Tf} \alpha \mu (\alpha + 3 \mu)^2 (-840 \mu^2 ((23 + 13 \gamma) \mu + 3 (5 + 2 \gamma) \nu) + \\
& \quad 56 \alpha \mu (-15 (11 + \gamma - 8 \gamma^2) \mu - x2^2 (23 + \gamma (-17 + 5 \gamma)) \mu - 45 (5 + 2 \gamma) \\
& \quad \nu + 3 x2^2 (-5 + 4 \gamma) \nu) + \alpha^2 ((3 x2^4 (85 + 7 (-2 + \gamma) \gamma) - \\
& \quad 28 x2^2 (59 + \gamma (19 + 29 \gamma)) + 14 (23 + \gamma (174 + 241 \gamma))) \mu - 3 \\
& \quad (28 x2^2 (5 - 4 \gamma) + x2^4 (-59 + 12 \gamma) + 14 (73 + 28 \gamma) \nu))) + \\
& \quad 16016 (2 - x1) \alpha \mu (\alpha + 3 \mu) (3360 \text{Tf} x2^4 \alpha^2 \gamma ((-5 + x2^2) \alpha - 10 \mu) \mu (\alpha + 3 \mu) - \\
& \quad 240 \text{Tf} x1 \alpha^2 \gamma \mu (\alpha + 3 \mu) \\
& \quad ((6 + 7 x2^4 (-5 + x2^2)) \alpha + 14 (1 - 5 x2^4) \mu) - \text{Tf}^2 x2^4 \alpha \\
& \quad (\alpha^3 (90 (-35 + \gamma) - 210 x2^2 (-2 + \gamma) + 45 x2^4 (3 + 2 \gamma) - 2 x2^6 (7 + 5 \gamma)) + \\
& \quad 30 \alpha^2 (7 (-90 + \gamma) - 28 x2^2 (-2 + \gamma) + x2^4 (9 + 6 \gamma)) \mu - \\
& \quad 840 \alpha (45 + x2^2 (-2 + \gamma)) \mu^2 - 25200 \mu^3) + 1680 (2 - x1) \alpha \mu (\alpha + 3 \mu)
\end{aligned}$$

$$\begin{aligned}
& (Tf \, x2^4 \alpha ((-5 + x2^2) \alpha - 10 \mu) + 40 (-2 + x1) \mu (\alpha - x2^2 \alpha + 2 \mu) (\alpha + 3 \mu)) - \\
& 5600 x2^4 \alpha^2 \mu (\alpha + 3 \mu)^2 ((8 + \gamma) \mu + 6 \nu) - \\
& 8400 \mu (\alpha - x2^2 \alpha + 2 \mu) (\alpha + 3 \mu) (3 Tf \, x1 (\alpha + 2 \mu)^2 + \\
& 4 (\alpha + 3 \mu) (\alpha (-5 + x2^2 + (-2 + x1)^2 \gamma) \mu + (-1 + x2^2) \alpha \nu - 2 \mu (\mu + \nu))) , \\
& 1 / (7884800 \gamma^3 (\alpha + 3 \mu)^6) \times 27 Tf^4
\end{aligned}$$

$$\alpha^4$$

$$3 \mu$$

$$\begin{aligned}
& (2640 Tf (2 - x1) x2 \alpha \gamma \mu (\alpha + 3 \mu) \\
& ((6 - 7 x2^4 + x2^6) \alpha - 14 (-1 + x2^4) \mu) - \\
& 5280 Tf \, x1 \alpha (2 + \gamma (2 + \gamma)) \mu (\alpha + 3 \mu) \\
& (x2^7 \alpha - 7 x2^5 (\alpha + 2 \mu) + 2 x2 (3 \alpha + 7 \mu)) + \\
& x2 (10560 Tf (-1 + x2^2) \alpha (2 + \gamma (2 + \gamma)) \mu (\alpha + 3 \mu) \\
& ((-6 - 6 x2^2 + x2^4) \alpha - 14 (1 + x2^2) \mu) - \\
& Tf^2 (\alpha^3 (12545 + 4 \gamma (4765 + 16 \gamma) + 55 x2^8 (17 + 4 \gamma (5 + \gamma)) + 198 \\
& x2^4 (-66 + \gamma (-103 + 2 \gamma)) - 330 x2^6 (1 + \gamma (-1 + 2 \gamma)) - 2 x2^{10} \\
& (41 + 2 \gamma (24 + 5 \gamma))) + 22 \alpha^2 (3671 + 5468 \gamma + 58 \gamma^2 + 42 x2^4 (-88 + \\
& (-134 + \gamma) \gamma) + 5 x2^8 (17 + 4 \gamma (5 + \gamma)) - 60 x2^6 (1 + \gamma (-1 + 2 \gamma))) \mu - \\
& 1320 \alpha (-127 - 2 \gamma (94 + \gamma) + 63 x2^4 (2 + 3 \gamma) + x2^6 (1 + \gamma (-1 + 2 \gamma))) \mu^2 - \\
& 55440 (-1 + x2^4) (2 + 3 \gamma) \mu^3) - \\
& 24640 (-1 + x2^4) \alpha (1 + \gamma) \mu (\alpha + 3 \mu)^2 ((8 + \gamma) \mu + 6 \nu))) ;
\end{aligned}$$

(*-----8th order-----*)

$$\begin{aligned}
\varphi 8 = & 1 / (31539200 \gamma^4 (\alpha + 3 \mu)^7) \times 27 Tf^4 \alpha \mu \\
& (-246400 (-2 + x1)^4 \alpha^3 (-3 + \gamma) (-2 + \gamma) (-1 + \gamma) \mu^3 (\alpha + 3 \mu)^3 + 44 x2^2 \alpha^2 \mu \\
& (\alpha + 3 \mu) (3360 Tf \, x2^2 \alpha (4 + \gamma (5 + 2 \gamma)) ((-5 + x2^2) \alpha - 10 \mu) \mu (\alpha + 3 \mu) - Tf^2 \\
& (\alpha^3 (630 x2^4 (1 + 2 \gamma) + 2 x2^8 (-1 + \gamma) (5 + 8 \gamma) + 180 x2^2 (-27 + 4 (-11 + \gamma) \gamma) + \\
& 45 x2^6 (1 - 4 \gamma^2) - 540 (4 + \gamma (5 + 2 \gamma))) - 30 \alpha^2 (-84 x2^4 (1 + 2 \gamma) + \\
& 7 x2^2 (164 + (253 - 8 \gamma) \gamma) + 3 x2^6 (-1 + 4 \gamma^2) + 78 (4 + \gamma (5 + 2 \gamma))) \mu + \\
& 2520 \alpha (-4 + x2^4 (1 + 2 \gamma) - \gamma (5 + 2 \gamma) - 15 x2^2 (2 + 3 \gamma)) \mu^2 - \\
& 25200 x2^2 (2 + 3 \gamma) \mu^3) - 11200 x2^2 \alpha (1 + \gamma) \mu (\alpha + 3 \mu)^2 ((8 + \gamma) \mu + 6 \nu)) + \\
& 369600 (-2 + x1)^2 \alpha^2 (1 - \gamma) (2 - \gamma) \mu^2 (\alpha + 3 \mu)^2 (3 Tf \, x1 (\alpha + 2 \mu)^2 + \\
& 4 (\alpha + 3 \mu) (\alpha (-5 + x2^2 + (-2 + x1)^2 \gamma) \mu + (-1 + x2^2) \alpha \nu - 2 \mu (\mu + \nu))) - \\
& x1 (739200 Tf \, x1^2 \alpha^2 (5 + 5 \gamma + \gamma^2) \mu^2 (\alpha + 2 \mu)^2 (\alpha + 3 \mu)^2 - \\
& 2217600 Tf \, x1 \alpha^2 (8 + 9 \gamma + 2 \gamma^2) \mu^2 (\alpha + 2 \mu)^2 (\alpha + 3 \mu)^2 - \\
& 1971200 x1^2 \alpha^3 (6 + 7 \gamma + 2 \gamma^2) \mu^3 (\alpha + 3 \mu)^3 + \\
& 246400 x1^3 \alpha^3 (6 + 7 \gamma + 2 \gamma^2) \mu^3 (\alpha + 3 \mu)^3 - \\
& 31680 Tf^2 \alpha \mu (\alpha^2 + 5 \alpha \mu + 6 \mu^2) (3 \alpha^3 (39 + 40 \gamma + 2 \gamma^2) + \\
& 7 \alpha^2 (94 + 95 \gamma + 2 \gamma^2) \mu + 1260 \alpha (1 + \gamma) \mu^2 + 840 (1 + \gamma) \mu^3) + \\
& 3960 Tf^2 x1 \alpha \mu (\alpha^2 + 5 \alpha \mu + 6 \mu^2) (3 \alpha^3 (183 + 121 \gamma + 4 \gamma^2) + \\
& 14 \alpha^2 (229 + 143 \gamma + 2 \gamma^2) \mu + 1260 \alpha (5 + 3 \gamma) \mu^2 + 840 (5 + 3 \gamma) \mu^3) + \\
& 3 Tf^3 (\alpha + 2 \mu) (\alpha^5 (116495 + 25000 \gamma + 64 \gamma^2) + 22 \alpha^4 (50921 + 7178 \gamma + 58 \gamma^2) \\
& \mu + 1320 \alpha^3 (3277 + 248 \gamma + 2 \gamma^2) \mu^2 + 221760 \alpha^2 (38 + \gamma) \mu^3 + \\
& 8316000 \alpha \mu^4 + 3326400 \mu^5) - 2956800 x1 \alpha^2 (2 + \gamma) \mu^2 (\alpha + 3 \mu)^3 \\
& (\alpha (-3 \gamma \mu + 2 \nu) + 2 \mu ((4 + \gamma) \mu + 2 \nu)) + 1971200 \alpha^2 \mu^2 (\alpha + 3 \mu)^3 \\
& (6 (3 + 2 \gamma) \mu ((4 + \gamma) \mu + 2 \nu) + \alpha ((32 + 21 \gamma - 2 \gamma^2) \mu + 6 (3 + 2 \gamma) \nu)) + \\
& 10560 Tf \alpha \mu (\alpha + 3 \mu)^2 (420 \alpha \mu^2 ((-23 + 6 \gamma + 8 \gamma^2) \mu - 3 (7 + 2 \gamma) \nu) - \\
& 840 \mu^3 ((13 + 6 \gamma) \mu + (7 + 2 \gamma) \nu) + \alpha^3 ((235 + 1791 \gamma + 821 \gamma^2 - 35 x2^4 \\
& (4 + 5 \gamma + 2 \gamma^2) + 7 x2^6 (4 + 5 \gamma + 2 \gamma^2)) \mu - 21 (37 + 12 \gamma) \nu) - 14 \alpha^2 \mu \\
& (5 x2^4 (4 + 5 \gamma + 2 \gamma^2) \mu - 3 (-39 + 145 \gamma + 79 \gamma^2) \mu + 3 (107 + 32 \gamma) \nu))) + \\
& 4 (\alpha + 3 \mu) (\mu + \nu) (2640 Tf \alpha \mu (\alpha + 3 \mu) (\alpha^3 (-28 (15 - 15 x2^2 - 5 x2^4 + x2^6) (1 + \gamma) +
\end{aligned}$$

$$\begin{aligned}
& x1 (420 + 321 \gamma - 70 x2^4 (1 + \gamma) + 14 x2^6 (1 + \gamma) - 105 x2^2 (4 + 3 \gamma)) - \\
& 28 \alpha^2 (-10 (-9 + 6 x2^2 + x2^4) (1 + \gamma) + x1 (-90 - 68 \gamma + \\
& \quad 5 x2^4 (1 + \gamma) + 15 x2^2 (4 + 3 \gamma))) \mu - 420 (-3 + x2^2) \alpha \\
& \quad (-4 (1 + \gamma) + x1 (4 + 3 \gamma)) \mu^2 + 840 (-4 (1 + \gamma) + x1 (4 + 3 \gamma)) \mu^3 - \\
& 11 Tf^2 (\alpha^5 (-90 x2^4 (-70 + \gamma) + 210 x2^6 (-5 + \gamma) - 45 x2^8 (3 + 2 \gamma) - 270 (35 + \\
& \quad 2 \gamma) + 270 x2^2 (35 + 2 \gamma) + 2 x2^{10} (7 + 5 \gamma)) - 30 \alpha^4 (7 x2^4 (-180 + \gamma) - \\
& \quad 28 x2^6 (-5 + \gamma) + x2^8 (9 + 6 \gamma) - 6 x2^2 (420 + 13 \gamma) + 6 (525 + 19 \gamma)) \mu + \\
& 120 \alpha^3 (630 x2^4 + 7 x2^6 (-5 + \gamma) + 21 x2^2 (90 + \gamma) - 30 (105 + 2 \gamma)) \mu^2 + \\
& 5040 \alpha^2 (-150 + 60 x2^2 + 10 x2^4 - \gamma) \mu^3 + 151200 (-5 + x2^2) \alpha \mu^4 - \\
& 302400 \mu^5) - 61600 \alpha \mu (\alpha + 3 \mu)^2 (24 \mu^2 ((7 + 3 \gamma) \mu + (5 + 2 \gamma) \nu) + \\
& \alpha^2 ((x2^4 (10 + 11 \gamma) - 24 x1 (-1 + x2^2) (6 + 7 \gamma + 2 \gamma^2) + \\
& \quad 6 x1^2 (-1 + x2^2) (6 + 7 \gamma + 2 \gamma^2) + 6 x2^2 (6 + 19 \gamma + 8 \gamma^2) - \\
& \quad 6 (9 + 21 \gamma + 8 \gamma^2)) \mu + 6 (5 + 2 \gamma + 2 x2^4 (1 + \gamma) - 4 x2^2 (2 + \gamma)) \nu) - \\
& 12 \alpha \mu ((2 + 18 \gamma + 8 \gamma^2 + 5 x2^2 (2 + \gamma) - 4 x1 (6 + 7 \gamma + 2 \gamma^2) + \\
& \quad x1^2 (6 + 7 \gamma + 2 \gamma^2)) \mu + 2 (-5 - 2 \gamma + 2 x2^2 (2 + \gamma)) \nu)) + \\
& 440 \alpha (1 - \gamma) \mu (\alpha + 3 \mu) (105 (3 Tf x1 (\alpha + 2 \mu)^2 + 4 (\alpha + 3 \mu) \\
& \quad (\alpha (-5 + x2^2 + (-2 + x1)^2 \gamma) \mu + (-1 + x2^2) \alpha \nu - 2 \mu (\mu + \nu)))^2 - 2 (2 - x1) \\
& \quad (9 Tf^2 x1 (\alpha + 2 \mu) (3 \alpha^3 (35 + 2 \gamma) + 14 \alpha^2 (45 + \gamma) \mu + 1260 \alpha \mu^2 + 840 \mu^3) + \\
& \quad 1120 \alpha \mu (\alpha + 3 \mu)^2 (\alpha (-2 - 6 x2^2 + 3 x1 (-5 + x2^2) + 36 \gamma + \\
& \quad x1 (-9 + (-6 + x1) x1) \gamma + 6 (-2 + x1) x2^2 \gamma + 2 (-2 + x1)^3 \gamma^2) \mu - \\
& \quad 6 (-2 - 4 \gamma + x1 (5 + 3 \gamma)) \mu^2 + 3 \alpha (2 - 3 x1 - 2 x2^2 + x1 x2^2 + 2 (-2 + x1) \\
& \quad (-1 + x2^2) \gamma) \nu - 6 (-2 + 3 x1 + 2 (-2 + x1) \gamma) \mu \nu) + 84 Tf (\alpha + 3 \mu) \\
& \quad (-120 \mu^3 (\mu + \nu) + 60 \alpha \mu^2 ((-3 + x2^2 + x1 (-8 + 3 x1) \gamma) \mu + (-3 + x2^2) \nu) + \\
& \quad 10 \alpha^2 \mu ((-9 + 6 x2^2 + x2^4 - (6 (8 - 3 x1) x1 + x2^4) \gamma) \mu + \\
& \quad (-9 + 6 x2^2 + x2^4) \nu) + \alpha^3 ((-15 + 15 x2^2 + x2^4 (-5 + x2^2) (-1 + \gamma) + \\
& \quad 15 x1 (-8 + 3 x1) \gamma) \mu - (15 - 15 x2^2 - 5 x2^4 + x2^6) \nu)))); \\
& u8 = \{1 / (22960537600 \gamma^4 (\alpha + 3 \mu)^8) \times 81 Tf^5 \alpha \mu \\
& \quad (-4004 \alpha (Tf x2^4 \alpha ((-5 + x2^2) \alpha - 10 \mu) + 40 (-2 + x1) \mu (\alpha - x2^2 \alpha + 2 \mu) (\alpha + 3 \mu)) \\
& \quad (-9 Tf^2 x1 (\alpha + 2 \mu) (3 \alpha^3 (35 + 2 \gamma) + 14 \alpha^2 (45 + \gamma) \mu + 1260 \alpha \mu^2 + 840 \mu^3) - \\
& \quad 1120 \alpha \mu (\alpha + 3 \mu)^2 (\alpha (-2 - 6 x2^2 + 3 x1 (-5 + x2^2) + 36 \gamma + x1 (-9 + (-6 + x1) x1) \\
& \quad \gamma + 6 (-2 + x1) x2^2 \gamma + 2 (-2 + x1)^3 \gamma^2) \mu - 6 (-2 - 4 \gamma + x1 (5 + 3 \gamma)) \mu^2 + \\
& \quad 3 \alpha (2 - 3 x1 - 2 x2^2 + x1 x2^2 + 2 (-2 + x1) (-1 + x2^2) \gamma) \nu - \\
& \quad 6 (-2 + 3 x1 + 2 (-2 + x1) \gamma) \mu \nu) - 84 Tf (\alpha + 3 \mu) (-120 \mu^3 (\mu + \nu) + \\
& \quad 60 \alpha \mu^2 ((-3 + x2^2 + x1 (-8 + 3 x1) \gamma) \mu + (-3 + x2^2) \nu) + 10 \alpha^2 \mu \\
& \quad ((-9 + 6 x2^2 + x2^4 - (6 (8 - 3 x1) x1 + x2^4) \gamma) \mu + (-9 + 6 x2^2 + x2^4) \nu) + \\
& \quad \alpha^3 ((-15 + 15 x2^2 + x2^4 (-5 + x2^2) (-1 + \gamma) + 15 x1 (-8 + 3 x1) \gamma) \\
& \quad \mu - (15 - 15 x2^2 - 5 x2^4 + x2^6) \nu)) + 2002 \alpha (3 Tf x1 (\alpha + 2 \mu)^2 + \\
& \quad 4 (\alpha + 3 \mu) (\alpha (-5 + x2^2 + (-2 + x1)^2 \gamma) \mu + (-1 + x2^2) \alpha \nu - 2 \mu (\mu + \nu))) \\
& \quad (-3360 Tf x2^4 \alpha^2 \gamma ((-5 + x2^2) \alpha - 10 \mu) \mu (\alpha + 3 \mu) + 240 Tf x1 \alpha^2 \gamma \mu \\
& \quad (\alpha + 3 \mu) ((6 + 7 x2^4 (-5 + x2^2)) \alpha + 14 (1 - 5 x2^4) \mu) + Tf^2 x2^4 \alpha \\
& \quad (\alpha^3 (90 (-35 + \gamma) - 210 x2^2 (-2 + \gamma) + 45 x2^4 (3 + 2 \gamma) - 2 x2^6 (7 + 5 \gamma)) + \\
& \quad 30 \alpha^2 (7 (-90 + \gamma) - 28 x2^2 (-2 + \gamma) + x2^4 (9 + 6 \gamma)) \mu - \\
& \quad 840 \alpha (45 + x2^2 (-2 + \gamma)) \mu^2 - 25200 \mu^3) - 1680 (2 - x1) \alpha \mu (\alpha + 3 \mu) \\
& \quad (Tf x2^4 \alpha ((-5 + x2^2) \alpha - 10 \mu) + 40 (-2 + x1) \mu (\alpha - x2^2 \alpha + 2 \mu) (\alpha + 3 \mu)) + \\
& \quad 5600 x2^4 \alpha^2 \mu (\alpha + 3 \mu)^2 ((8 + \gamma) \mu + 6 \nu) + \\
& \quad 8400 \mu (\alpha - x2^2 \alpha + 2 \mu) (\alpha + 3 \mu) (3 Tf x1 (\alpha + 2 \mu)^2 + \\
& \quad 4 (\alpha + 3 \mu) (\alpha (-5 + x2^2 + (-2 + x1)^2 \gamma) \mu + (-1 + x2^2) \alpha \nu - 2 \mu (\mu + \nu))) - \\
& \quad (2 - x1) \alpha (3843840 Tf x1^2 \alpha^3 (2 + 2 \gamma + \gamma^2) \mu^2 (\alpha + 3 \mu)^2 \\
& \quad ((6 - 35 x2^4 + 7 x2^6) \alpha + 14 (1 - 5 x2^4) \mu) +
\end{aligned}$$

$$\begin{aligned}
& 3363360 \alpha \mu (\alpha + 3 \mu) (\text{Tf } x2^4 \alpha ((-5 + x2^2) \alpha - 10 \mu) + \\
& 40 (-2 + x1) \mu (\alpha - x2^2 \alpha + 2 \mu) (\alpha + 3 \mu)) (3 \text{Tf } x1 (\alpha + 2 \mu)^2 + \\
& 4 (\alpha + 3 \mu) (\alpha (-5 + x2^2 + (-2 + x1)^2 \gamma) \mu + (-1 + x2^2) \alpha \nu - 2 \mu (\mu + \nu))) - \\
& 1456 x1 \alpha^2 \mu (\alpha + 3 \mu) (10560 \text{Tf} (-1 + x2^2) \alpha (2 + \gamma (2 + \gamma)) \mu \\
& (\alpha + 3 \mu) ((-6 - 6 x2^2 + x2^4) \alpha - 14 (1 + x2^2) \mu) - \\
& \text{Tf}^2 (\alpha^3 (12545 + 4 \gamma (4765 + 16 \gamma) + 55 x2^8 (17 + 4 \gamma (5 + \gamma)) + \\
& 198 x2^4 (-66 + \gamma (-103 + 2 \gamma)) - 330 x2^6 (1 + \gamma (-1 + 2 \gamma)) - \\
& 2 x2^{10} (41 + 2 \gamma (24 + 5 \gamma))) + 22 \alpha^2 (3671 + 5468 \gamma + 58 \gamma^2 + \\
& 42 x2^4 (-88 + (-134 + \gamma) \gamma) + 5 x2^8 (17 + 4 \gamma (5 + \gamma)) - 60 x2^6 \\
& (1 + \gamma (-1 + 2 \gamma))) \mu - 1320 \alpha (-127 - 2 \gamma (94 + \gamma) + 63 x2^4 (2 + 3 \gamma) + \\
& x2^6 (1 + \gamma (-1 + 2 \gamma))) \mu^2 - 55440 (-1 + x2^4) (2 + 3 \gamma) \mu^3) - \\
& 24640 (-1 + x2^4) \alpha (1 + \gamma) \mu (\alpha + 3 \mu)^2 ((8 + \gamma) \mu + 6 \nu) + \\
& x2 (42240 \text{Tf } x2 (-1 + x2^2) \alpha (2 + 2 \gamma + \gamma^2) ((-3 + x2^2) \alpha - 7 \mu) \mu \\
& (\alpha + 3 \mu) + 21120 \text{Tf } x2 \alpha (2 + \gamma (2 + \gamma)) \mu (\alpha + 3 \mu) ((-6 - 6 x2^2 + x2^4) \alpha - \\
& 14 (1 + x2^2) \mu) - 4 \text{Tf}^2 x2^3 (-5 x2^6 \alpha^3 (41 + 48 \gamma + 10 \gamma^2) + \\
& 110 x2^4 \alpha^2 (17 + 20 \gamma + 4 \gamma^2) (\alpha + 2 \mu) - 495 x2^2 \alpha (1 - \gamma + 2 \gamma^2) \\
& (\alpha + 2 \mu)^2 + 66 (3 \alpha^3 (-66 - 103 \gamma + 2 \gamma^2) + 14 \alpha^2 (-88 - 134 \gamma + \gamma^2) \\
& \mu - 1260 \alpha (2 + 3 \gamma) \mu^2 - 840 (2 + 3 \gamma) \mu^3)) - 98560 x2^3 \alpha (1 + \gamma) \mu \\
& (\alpha + 3 \mu)^2 ((8 + \gamma) \mu + 6 \nu))) + 160160 \mu (\alpha - x2^2 \alpha + 2 \mu) (\alpha + 3 \mu) \\
& (9 \text{Tf}^2 x1 (\alpha + 2 \mu) (3 \alpha^3 (35 + 2 \gamma) + 14 \alpha^2 (45 + \gamma) \mu + 1260 \alpha \mu^2 + 840 \mu^3) + \\
& 1120 \alpha \mu (\alpha + 3 \mu)^2 (\alpha (-2 - 6 x2^2 + 3 x1 (-5 + x2^2) + 36 \gamma + \\
& x1 (-9 + (-6 + x1) x1) \gamma + 6 (-2 + x1) x2^2 \gamma + 2 (-2 + x1)^3 \gamma^2) \mu - 6 \\
& (-2 - 4 \gamma + x1 (5 + 3 \gamma)) \mu^2 + 3 \alpha (2 - 3 x1 - 2 x2^2 + x1 x2^2 + 2 (-2 + x1) \\
& (-1 + x2^2) \gamma) \nu - 6 (-2 + 3 x1 + 2 (-2 + x1) \gamma) \mu \nu) + 84 \text{Tf} (\alpha + 3 \mu) \\
& (-120 \mu^3 (\mu + \nu) + 60 \alpha \mu^2 ((-3 + x2^2 + x1 (-8 + 3 x1) \gamma) \mu + (-3 + x2^2) \nu) + \\
& 10 \alpha^2 \mu ((-9 + 6 x2^2 + x2^4 - (6 (8 - 3 x1) x1 + x2^4) \gamma) \mu + (-9 + 6 x2^2 + \\
& x2^4) \nu) + \alpha^3 ((-15 + 15 x2^2 + x2^4 (-5 + x2^2) (-1 + \gamma) + \\
& 15 x1 (-8 + 3 x1) \gamma) \mu - (15 - 15 x2^2 - 5 x2^4 + x2^6) \nu))) + x2^4 \alpha \\
& (32032 \text{Tf}^2 \alpha \mu (\alpha + 3 \mu) (\alpha^3 (-90 (-31 + 2 (-33 + \gamma) \gamma) + 420 x2^2 (2 + \gamma + \\
& \gamma^2) - 45 x2^4 (17 + 4 \gamma (5 + \gamma)) + x2^6 (82 + 4 \gamma (24 + 5 \gamma))) - 30 \alpha^2 \\
& (14 (-43 + (-88 + \gamma) \gamma) - 56 x2^2 (2 + \gamma + \gamma^2) + 3 x2^4 (17 + 4 \gamma (5 + \gamma))) \\
& \mu + 840 \alpha (45 + 90 \gamma + 2 x2^2 (2 + \gamma + \gamma^2)) \mu^2 + 25200 (1 + 2 \gamma) \mu^3) + \\
& \text{Tf}^3 (\alpha^5 (-91 x2^8 (15 + 8 \gamma) (93 + 20 \gamma) + 91 (-195355 + 7180 \gamma + 64 \gamma^2) - \\
& 1287 x2^4 (-3009 - 832 \gamma + 88 \gamma^2) + 12 x2^{10} (754 + 575 \gamma + 90 \gamma^2) + \\
& 36036 x2^2 (-101 + \gamma (-89 + 2 \gamma)) + 2002 x2^6 (9 + 8 \gamma (13 + 4 \gamma))) - \\
& 182 \alpha^4 (999119 - 22528 \gamma - 638 \gamma^2 + x2^8 (15 + 8 \gamma) (93 + 20 \gamma) + \\
& 132 x2^2 (1234 + (1156 - 13 \gamma) \gamma) - 44 x2^6 (9 + 8 \gamma (13 + 4 \gamma)) + \\
& 99 x2^4 (-1299 + \gamma (-377 + 33 \gamma))) \mu + 4004 \alpha^3 \\
& (84 x2^2 (-268 + (-262 + \gamma) \gamma) - 135 x2^4 (-87 + 2 (-13 + \gamma) \gamma) + \\
& 30 (-6173 + 2 \gamma (34 + \gamma)) + 2 x2^6 (9 + 8 \gamma (13 + 4 \gamma))) \mu^2 - 360360 \alpha^2 \\
& (-14 (-298 + \gamma) + 336 x2^2 (1 + \gamma) + x2^4 (-87 + 2 (-13 + \gamma) \gamma)) \\
& \mu^3 - 60540480 \alpha (25 + x2^2 (1 + \gamma)) \mu^4 - 605404800 \mu^5) - \\
& 358758400 \alpha^2 (1 + \gamma) \mu^2 (\alpha + 3 \mu)^3 ((8 + \gamma) \mu + 6 \nu) - \\
& 160160 \text{Tf} \alpha \mu (\alpha + 3 \mu)^2 (-840 \mu^2 ((23 + 13 \gamma) \mu + 3 (5 + 2 \gamma) \nu) + 56 \alpha \mu \\
& (-15 (11 + \gamma - 8 \gamma^2) \mu - x2^2 (23 + \gamma (-17 + 5 \gamma)) \mu - 45 (5 + 2 \gamma) \nu + \\
& 3 x2^2 (-5 + 4 \gamma) \nu) + \alpha^2 ((3 x2^4 (85 + 7 (-2 + \gamma) \gamma) - \\
& 28 x2^2 (59 + \gamma (19 + 29 \gamma)) + 14 (23 + \gamma (174 + 241 \gamma))) \mu - \\
& 3 (28 x2^2 (5 - 4 \gamma) + x2^4 (-59 + 12 \gamma) + 14 (73 + 28 \gamma)) \nu))) + \\
& 16016 (2 - x1) \alpha \mu (\alpha + 3 \mu) (3360 \text{Tf } x2^4 \alpha^2 \gamma ((-5 + x2^2) \alpha - 10 \mu) \mu (\alpha + 3 \mu) - \\
& 240 \text{Tf } x1 \alpha^2 \gamma \mu (\alpha + 3 \mu) ((6 + 7 x2^4 (-5 + x2^2)) \alpha + 14 (1 - 5 x2^4) \mu) -
\end{aligned}$$

$$\begin{aligned}
& \text{Tf}^2 x^2^4 \alpha (\alpha^3 (90 (-35 + \gamma) - 210 x^2^2 (-2 + \gamma) + 45 x^2^4 (3 + 2 \gamma) - \\
& \quad 2 x^2^6 (7 + 5 \gamma)) + 30 \alpha^2 (7 (-90 + \gamma) - 28 x^2^2 (-2 + \gamma) + x^2^4 (9 + 6 \gamma)) \\
& \quad \mu - 840 \alpha (45 + x^2^2 (-2 + \gamma)) \mu^2 - 25200 \mu^3) + 1680 (2 - x1) \alpha \mu (\alpha + 3 \mu) \\
& \quad (\text{Tf} x^2^4 \alpha ((-5 + x^2^2) \alpha - 10 \mu) + 40 (-2 + x1) \mu (\alpha - x^2^2 \alpha + 2 \mu) (\alpha + 3 \mu)) - \\
& \quad 5600 x^2^4 \alpha^2 \mu (\alpha + 3 \mu)^2 ((8 + \gamma) \mu + 6 \nu) - \\
& \quad 8400 \mu (\alpha - x^2^2 \alpha + 2 \mu) (\alpha + 3 \mu) (3 \text{Tf} x1 (\alpha + 2 \mu)^2 + 4 (\alpha + 3 \mu) \\
& \quad (\alpha (-5 + x^2^2 + (-2 + x1)^2 \gamma) \mu + (-1 + x^2^2) \alpha \nu - 2 \mu (\mu + \nu)))) - \\
182 & (\alpha - x^2^2 \alpha + 2 \mu) (-246400 (-2 + x1)^4 \alpha^3 (-3 + \gamma) (-2 + \gamma) (-1 + \gamma) \\
& \mu^3 (\alpha + 3 \mu)^3 + 44 x^2^2 \alpha^2 \mu (\alpha + 3 \mu) \\
& (3360 \text{Tf} x^2^2 \alpha (4 + \gamma (5 + 2 \gamma)) ((-5 + x^2^2) \alpha - 10 \mu) \mu (\alpha + 3 \mu) - \\
& \text{Tf}^2 (\alpha^3 (630 x^2^4 (1 + 2 \gamma) + 2 x^2^8 (-1 + \gamma) (5 + 8 \gamma) + 180 x^2^2 \\
& \quad (-27 + 4 (-11 + \gamma) \gamma) + 45 x^2^6 (1 - 4 \gamma^2) - 540 (4 + \gamma (5 + 2 \gamma))) - \\
& \quad 30 \alpha^2 (-84 x^2^4 (1 + 2 \gamma) + 7 x^2^2 (164 + (253 - 8 \gamma) \gamma) + \\
& \quad 3 x^2^6 (-1 + 4 \gamma^2) + 78 (4 + \gamma (5 + 2 \gamma))) \mu + 2520 \alpha (-4 + x^2^4 (1 + 2 \gamma) - \\
& \quad \gamma (5 + 2 \gamma) - 15 x^2^2 (2 + 3 \gamma)) \mu^2 - 25200 x^2^2 (2 + 3 \gamma) \mu^3) - \\
& \quad 11200 x^2^2 \alpha (1 + \gamma) \mu (\alpha + 3 \mu)^2 ((8 + \gamma) \mu + 6 \nu) + 369600 \\
& (-2 + x1)^2 \alpha^2 (1 - \gamma) (2 - \gamma) \mu^2 (\alpha + 3 \mu)^2 (3 \text{Tf} x1 (\alpha + 2 \mu)^2 + \\
& \quad 4 (\alpha + 3 \mu) (\alpha (-5 + x^2^2 + (-2 + x1)^2 \gamma) \mu + (-1 + x^2^2) \alpha \nu - 2 \mu (\mu + \nu))) - \\
x1 & (739200 \text{Tf} x1^2 \alpha^2 (5 + 5 \gamma + \gamma^2) \mu^2 (\alpha + 2 \mu)^2 (\alpha + 3 \mu)^2 - \\
& \quad 2217600 \text{Tf} x1 \alpha^2 (8 + 9 \gamma + 2 \gamma^2) \mu^2 (\alpha + 2 \mu)^2 (\alpha + 3 \mu)^2 - \\
& \quad 1971200 x1^2 \alpha^3 (6 + 7 \gamma + 2 \gamma^2) \mu^3 (\alpha + 3 \mu)^3 + \\
& \quad 246400 x1^3 \alpha^3 (6 + 7 \gamma + 2 \gamma^2) \mu^3 (\alpha + 3 \mu)^3 - \\
& \quad 31680 \text{Tf}^2 \alpha \mu (\alpha^2 + 5 \alpha \mu + 6 \mu^2) (3 \alpha^3 (39 + 40 \gamma + 2 \gamma^2) + 7 \\
& \quad \alpha^2 (94 + 95 \gamma + 2 \gamma^2) \mu + 1260 \alpha (1 + \gamma) \mu^2 + 840 (1 + \gamma) \mu^3) + \\
& \quad 3960 \text{Tf}^2 x1 \alpha \mu (\alpha^2 + 5 \alpha \mu + 6 \mu^2) (3 \alpha^3 (183 + 121 \gamma + 4 \gamma^2) + 14 \\
& \quad \alpha^2 (229 + 143 \gamma + 2 \gamma^2) \mu + 1260 \alpha (5 + 3 \gamma) \mu^2 + 840 (5 + 3 \gamma) \mu^3) + \\
& \quad 3 \text{Tf}^3 (\alpha + 2 \mu) (\alpha^5 (116495 + 25000 \gamma + 64 \gamma^2) + 22 \alpha^4 \\
& \quad (50921 + 7178 \gamma + 58 \gamma^2) \mu + 1320 \alpha^3 (3277 + 248 \gamma + 2 \gamma^2) \\
& \quad \mu^2 + 221760 \alpha^2 (38 + \gamma) \mu^3 + 8316000 \alpha \mu^4 + 3326400 \mu^5) - \\
& \quad 2956800 x1 \alpha^2 (2 + \gamma) \mu^2 (\alpha + 3 \mu)^3 (\alpha (-3 \gamma \mu + 2 \nu) + 2 \mu ((4 + \gamma) \mu + 2 \nu)) + \\
& \quad 1971200 \alpha^2 \mu^2 (\alpha + 3 \mu)^3 \\
& \quad (6 (3 + 2 \gamma) \mu ((4 + \gamma) \mu + 2 \nu) + \alpha ((32 + 21 \gamma - 2 \gamma^2) \mu + 6 (3 + 2 \gamma) \nu)) + \\
& \quad 10560 \text{Tf} \alpha \mu (\alpha + 3 \mu)^2 (420 \alpha \mu^2 ((-23 + 6 \gamma + 8 \gamma^2) \mu - 3 (7 + 2 \gamma) \nu) - \\
& \quad 840 \mu^3 ((13 + 6 \gamma) \mu + (7 + 2 \gamma) \nu) + \alpha^3 \\
& \quad ((235 + 1791 \gamma + 821 \gamma^2 - 35 x^2^4 (4 + 5 \gamma + 2 \gamma^2) + 7 x^2^6 (4 + 5 \gamma + 2 \gamma^2)) \\
& \quad \mu - 21 (37 + 12 \gamma) \nu) - 14 \alpha^2 \mu (5 x^2^4 (4 + 5 \gamma + 2 \gamma^2) \mu - \\
& \quad 3 (-39 + 145 \gamma + 79 \gamma^2) \mu + 3 (107 + 32 \gamma) \nu))) + 4 (\alpha + 3 \mu) (\mu + \nu) \\
(2640 & \text{Tf} \alpha \mu (\alpha + 3 \mu) (\alpha^3 (-28 (15 - 15 x^2^2 - 5 x^2^4 + x^2^6) (1 + \gamma) + \\
& \quad x1 (420 + 321 \gamma - 70 x^2^4 (1 + \gamma) + 14 x^2^6 (1 + \gamma) - 105 x^2^2 (4 + 3 \gamma))) - \\
& \quad 28 \alpha^2 (-10 (-9 + 6 x^2^2 + x^2^4) (1 + \gamma) + x1 (-90 - 68 \gamma + \\
& \quad 5 x^2^4 (1 + \gamma) + 15 x^2^2 (4 + 3 \gamma))) \mu - 420 (-3 + x^2^2) \alpha \\
& \quad (-4 (1 + \gamma) + x1 (4 + 3 \gamma)) \mu^2 + 840 (-4 (1 + \gamma) + x1 (4 + 3 \gamma)) \mu^3) - \\
11 & \text{Tf}^2 (\alpha^5 (-90 x^2^4 (-70 + \gamma) + 210 x^2^6 (-5 + \gamma) - 45 x^2^8 (3 + 2 \gamma) - \\
& \quad 270 (35 + 2 \gamma) + 270 x^2^2 (35 + 2 \gamma) + 2 x^2^{10} (7 + 5 \gamma)) - 30 \\
& \quad \alpha^4 (7 x^2^4 (-180 + \gamma) - 28 x^2^6 (-5 + \gamma) + x^2^8 (9 + 6 \gamma) - \\
& \quad 6 x^2^2 (420 + 13 \gamma) + 6 (525 + 19 \gamma)) \mu + 120 \alpha^3 (630 x^2^4 + \\
& \quad 7 x^2^6 (-5 + \gamma) + 21 x^2^2 (90 + \gamma) - 30 (105 + 2 \gamma)) \mu^2 + 5040 \alpha^2 (-150 + \\
& \quad 60 x^2^2 + 10 x^2^4 - \gamma) \mu^3 + 151200 (-5 + x^2^2) \alpha \mu^4 - 302400 \mu^5) - \\
61600 & \alpha \mu (\alpha + 3 \mu)^2 (24 \mu^2 ((7 + 3 \gamma) \mu + (5 + 2 \gamma) \nu) + \alpha^2 \\
& \quad ((x^2^4 (10 + 11 \gamma) - 24 x1 (-1 + x^2^2) (6 + 7 \gamma + 2 \gamma^2) + 6 x1^2 (-1 + x^2^2)
\end{aligned}$$

```

(6 + 7 γ + 2 γ^2) + 6 x2^2 (6 + 19 γ + 8 γ^2) - 6 (9 + 21 γ + 8 γ^2) ) μ +
6 (5 + 2 γ + 2 x2^4 (1 + γ) - 4 x2^2 (2 + γ) ) ν) - 12 α μ
((2 + 18 γ + 8 γ^2 + 5 x2^2 (2 + γ) - 4 x1 (6 + 7 γ + 2 γ^2) +
x1^2 (6 + 7 γ + 2 γ^2) ) μ + 2 (-5 - 2 γ + 2 x2^2 (2 + γ) ) ν) ) +
440 α (1 - γ) μ (α + 3 μ) (105 (3 Tf x1 (α + 2 μ) ^2 + 4 (α + 3 μ)
(α (-5 + x2^2 + (-2 + x1) ^2 γ) μ + (-1 + x2^2) α ν - 2 μ (μ + ν) ) ) ^2 -
2 (2 - x1) (9 Tf^2 x1 (α + 2 μ) (3 α^3 (35 + 2 γ) + 14 α^2 (45 + γ) μ +
1260 α μ^2 + 840 μ^3) + 1120 α μ (α + 3 μ) ^2
(α (-2 - 6 x2^2 + 3 x1 (-5 + x2^2) + 36 γ + x1 (-9 + (-6 + x1) x1) γ +
6 (-2 + x1) x2^2 γ + 2 (-2 + x1) ^3 γ^2) μ - 6 (-2 - 4 γ + x1 (5 + 3 γ) )
μ^2 + 3 α (2 - 3 x1 - 2 x2^2 + x1 x2^2 + 2 (-2 + x1) (-1 + x2^2) γ) ν -
6 (-2 + 3 x1 + 2 (-2 + x1) γ) μ ν) + 84 Tf (α + 3 μ) (-120 μ^3 (μ + ν) +
60 α μ^2 ((-3 + x2^2 + x1 (-8 + 3 x1) γ) μ + (-3 + x2^2) ν) + 10 α^2 μ
((-9 + 6 x2^2 + x2^4 - (6 (8 - 3 x1) x1 + x2^4) γ) μ + (-9 + 6 x2^2 +
x2^4) ν) + α^3 ((-15 + 15 x2^2 + x2^4 (-5 + x2^2) (-1 + γ) +
15 x1 (-8 + 3 x1) γ) μ - (15 - 15 x2^2 - 5 x2^4 + x2^6) ν) ) ) ) ) ,
1 / (15 769 600 γ^4 (α + 3 μ) ^7) × 81 Tf^5 α^4 μ^2
(-330
Tf
x2
γ
((6 - 7 x2^4 + x2^6) α -
14 (-1 + x2^4) μ) (3 Tf x1 (α + 2 μ) ^2 +
4 (α + 3 μ) (α (-5 + x2^2 + (-2 + x1) ^2 γ) μ + (-1 + x2^2) α ν - 2 μ (μ + ν) ) ) -
(2 - x1) (2640 Tf (2 - x1) x2 α γ μ (α + 3 μ) ((6 - 7 x2^4 + x2^6) α - 14 (-1 + x2^4) μ) -
5280 Tf x1 α (2 + γ (2 + γ) ) μ (α + 3 μ)
(x2^7 α - 7 x2^5 (α + 2 μ) + 2 x2 (3 α + 7 μ) ) +
x2 (10560 Tf (-1 + x2^2) α (2 + γ (2 + γ) ) μ (α + 3 μ)
((-6 - 6 x2^2 + x2^4) α - 14 (1 + x2^2) μ) -
Tf^2 (α^3 (12545 + 4 γ (4765 + 16 γ) + 55 x2^8 (17 + 4 γ (5 + γ) ) +
198 x2^4 (-66 + γ (-103 + 2 γ) ) - 330 x2^6 (1 + γ (-1 + 2 γ) ) -
2 x2^10 (41 + 2 γ (24 + 5 γ) ) ) + 22 α^2 (3671 + 5468 γ + 58 γ^2 +
42 x2^4 (-88 + (-134 + γ) γ) + 5 x2^8 (17 + 4 γ (5 + γ) ) - 60 x2^6
(1 + γ (-1 + 2 γ) ) ) μ - 1320 α (-127 - 2 γ (94 + γ) + 63 x2^4 (2 + 3 γ) +
x2^6 (1 + γ (-1 + 2 γ) ) ) μ^2 - 55440 (-1 + x2^4) (2 + 3 γ) μ^3) -
24640 (-1 + x2^4) α (1 + γ) μ (α + 3 μ) ^2 ((8 + γ) μ + 6 ν) ) ) } ;

(* =====)
(*Verification of all equations (1--8)+boundary conditions*)
(* =====)
startTime = AbsoluteTime[];

(*Equation 1*)
eq1 = Grad[φ2, {x1, x2}] - μ / γ Laplacian[u0, {x1, x2}] // FullSimplify;
test1 = (eq1 == {0, 0});

(*Equation 2*)
eq2 = Div[u2, {x1, x2}] + Grad[φ2, {x1, x2}].u0 // FullSimplify;
test2 = (eq2 == 0);

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```

(*Equation 3*)
eq3 = Grad[φ4, {x1, x2}] - μ / γ Laplacian[u2, {x1, x2}] -
      (μ + ν) / γ Grad[Div[u2, {x1, x2}], {x1, x2}] + 1 / γ Grad[u0, {x1, x2}].u2 +
      1 / γ Grad[u2, {x1, x2}].u0 + (γ - 1) φ2 Grad[φ2, {x1, x2}] // FullSimplify;
test3 = (eq3 == {0, 0});

(*Equation 4*)
eq4 = Div[u4, {x1, x2}] + Grad[φ2, {x1, x2}].u2 +
      Grad[φ4, {x1, x2}].u0 + φ2 Div[u2, {x1, x2}] // FullSimplify;
test4 = (eq4 == 0);

(*Equation 5*)
eq5 = Grad[φ6, {x1, x2}] - μ / γ Laplacian[u4, {x1, x2}] -
      (μ + ν) / γ Grad[Div[u4, {x1, x2}], {x1, x2}] + 1 / γ Grad[u0, {x1, x2}].u4 +
      1 / γ Grad[u4, {x1, x2}].u0 + 1 / γ φ2 Grad[u0, {x1, x2}].u2 +
      1 / γ φ2 Grad[u2, {x1, x2}].u0 + ((γ - 1) (γ - 2)) / 2 φ2^2 Grad[φ2, {x1, x2}] +
      (γ - 1) (φ2 Grad[φ4, {x1, x2}] + φ4 Grad[φ2, {x1, x2}]) // FullSimplify;
test5 = (eq5 == {0, 0});

(*Equation 6*)
eq6 = Div[u6, {x1, x2}] + Grad[φ2, {x1, x2}].u4 + Grad[φ4, {x1, x2}].u2 +
      Grad[φ6, {x1, x2}].u0 + φ4 Div[u2, {x1, x2}] + φ2 Div[u4, {x1, x2}] // FullSimplify;
test6 = (eq6 == 0);

(*Equation 7 (momentum for φ8)*)
eq7 = Grad[φ8, {x1, x2}] - μ / γ Laplacian[u6, {x1, x2}] -
      (μ + ν) / γ Grad[Div[u6, {x1, x2}], {x1, x2}] +
      1 / γ (Grad[u6, {x1, x2}].u0 + Grad[u0, {x1, x2}].u6) +
      1 / γ (Grad[u4, {x1, x2}].u2 + Grad[u2, {x1, x2}].u4) +
      1 / γ φ2 (Grad[u4, {x1, x2}].u0 + Grad[u2, {x1, x2}].u2 + Grad[u0, {x1, x2}].u4) +
      1 / γ φ4 (Grad[u2, {x1, x2}].u0 + Grad[u0, {x1, x2}].u2) +
      (γ - 1) (φ2 Grad[φ6, {x1, x2}] + φ4 Grad[φ4, {x1, x2}] + φ6 Grad[φ2, {x1, x2}]) +
      ((γ - 1) (γ - 2)) / 2 (φ2^2 Grad[φ4, {x1, x2}] + 2 φ2 φ4 Grad[φ2, {x1, x2}]) +
      ((γ - 1) (γ - 2) (γ - 3)) / 6 φ2^3 Grad[φ2, {x1, x2}] // FullSimplify;
test7 = (eq7 == {0, 0});

(*Equation 8 (continuity for u8)*)
eq8 = Div[u8, {x1, x2}] + Grad[φ2, {x1, x2}].u6 + Grad[φ4, {x1, x2}].u4 +
      Grad[φ6, {x1, x2}].u2 + Grad[φ8, {x1, x2}].u0 + φ6 Div[u2, {x1, x2}] +
      φ4 Div[u4, {x1, x2}] + φ2 Div[u6, {x1, x2}] // FullSimplify;
test8 = (eq8 == 0);

endTime = AbsoluteTime[];
totalTime = endTime - startTime;

(*-----Output-----*)
Print["====="];
Print["Verification up to 8th order (x1,x2)"];
Print["====="];
Print["eq1: ", test1];

```



```

Print["eq2: ", test2];
Print["eq3: ", test3];
Print["eq4: ", test4];
Print["eq5: ", test5];
Print["eq6: ", test6];
Print["eq7: ", test7];
Print["eq8: ", test8];
Print["All main equations passed: ",
      test1 && test2 && test3 && test4 && test5 && test6 && test7 && test8];

(*Boundary checks for y-components at x2=±1*)
b2 = {FullSimplify[u2[[2]] /. x2 → 1] == 0, FullSimplify[u2[[2]] /. x2 → -1] == 0};
b4 = {FullSimplify[u4[[2]] /. x2 → 1] == 0, FullSimplify[u4[[2]] /. x2 → -1] == 0};
b6 = {FullSimplify[u6[[2]] /. x2 → 1] == 0, FullSimplify[u6[[2]] /. x2 → -1] == 0};
b8 = {FullSimplify[u8[[2]] /. x2 → 1] == 0, FullSimplify[u8[[2]] /. x2 → -1] == 0};
allBdry = And@@Flatten[{b2, b4, b6, b8}];

Print["-----"];
Print["Boundary conditions (v = 0 at x2 = ±1):"];
Print[" u2: ", b2];
Print[" u4: ", b4];
Print[" u6: ", b6];
Print[" u8: ", b8];
Print["All boundary tests passed: ", allBdry];

Print["-----"];
Print["Overall status: ",
      test1 && test2 && test3 && test4 && test5 && test6 && test7 && test8 && allBdry];
Print["Total computation time: ", totalTime, " seconds"];
Print["====="];

=====
Verification up to 8th order (x1,x2)

=====
eq1: True
eq2: True
eq3: True
eq4: True
eq5: True
eq6: True
eq7: True
eq8: True
All main equations passed: True

-----
Boundary conditions (v = 0 at x2 = ±1):
u2: {True, True}
u4: {True, True}

```

u6: {True, True}

u8: {True, True}

All boundary tests passed: True

Overall status: True

Total computation time: 0.3311341 seconds

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